

Liquid-Liquid Extraction

Final Lab Report

Unit Operations Lab 2

February 17, 2020

Group #4, Tuesday, PM section

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1. Executive Summary (10 pts)

The objective of this experiment is to extract acetic acid (0.5 wt%) from Keller-Heartt white oil by using water as the solvent and examine the efficiency of a three-stage-mixer-settler system at different impeller speeds. In this study, the independent variable is impeller speed of the agitator (400, 600, and 900 RPM) and the dependent variables are the raffinate and extract concentrations. As the stage number increased, the acetic acid concentration decreased in the oil, whereas the acetic acid concentration increased in the water. This was also observed for increasing impeller speed and can be observed in the results section (Figures 1-3). Using the McCabe Thiele diagrams (Figures 4-6), the theoretical number of stages are found to be 2.6, 3.2, and 2.8 for 400 RPM, 600 RPM, and 900 RPM, respectively. The overall efficiencies were found to be 86.6 %, 106.7 %, and 93.3 % for 400, 600, and 900 RPM, respectively. The calculated Murphee efficiency for an extract phase for stage 1 is found to be 85.24 % and for the raffinate phase for stage 1 is found to be 60.01 %.

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2. Introduction (10 pts)

Liquid-liquid extraction, also known as partitioning, is a practice of separating immiscible fluids or partially miscible by transferring a solute from one solvent to another. Water is commonly used as the solvent, while the other liquid may be a nonpolar organic liquid, however, the solvent is chosen based on its polarity since polar solvents can extract polar analytes more efficiently than the non-polar ones [1].

The purpose of this experiment is to operate a three-stage mixer-settler system in order to extract acetic acid from Keller-Heartt white oil by using water as the solvent. While operating the system, the effects of the speed of the impellers mixing the oil mixture and the efficiency of the extraction must be observed and analyzed.

In the feed tank, 25 gallons of pure white oil was mixed with 387mL of acetic acid in order to create a mixture of 0.5 wt% of acetic acid. The mixture was recirculated within the feed tank to ensure a proper mixing while the three-stage tanks were being filled with the water. After the system was set up, the speed of the three agitators was set to 400 RPM and the flow of the feed and water were also adjusted using the rotameter to 0.448 GPM and 0.784 GPM, respectively. As the oil passed through the stages, it interacted with water in a countercurrent flow, allowing the acetic acid to be extracted by the water. The system reached steady state prior to sample collection.

Two samples were collected from each stage tank, which were composed of oil sample and water sample and were titrated with $\frac{N}{10}$ sodium hydroxide (NaOH) to determine the amount of acetic acid present in either the extract (water) and the raffinate (oil) sample. After the six samples were collected, the procedure was repeated for 600 RPM and 900 RPM agitator speed. A total of 18 samples were collected during this experiment and were submitted to titration.

Commented [PTH3]: this statement is not right. Stage 3 will always have the least amount of acetic acid because it's the first stage of water and the last stage of oil. So by the time the oil get to the third stage, the acetic acid concentration is going to be low. So as the stage number increase (going from feed to product tank) the acetic acid decrease in water and oil.

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Liquid-liquid extraction is a common method used in research and chemical analysis due to its inexpensive and easy method of implementation [1]. This practice is commonly used in the industrial application, such as in the “fermentation products (antibiotics, amino acids, steroids)” [1] in the production of pharmaceuticals and the mineral industry.

3. Theory (10 pts)

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The application of polarity gives rise to the scheme of liquid – liquid extraction. In essence, the liquid polar structures vary to that of non-polar components which govern solubility and separation. Both the raffinate and extraction phases contain mole fractions of the solvent. It is expected as the stages of separation increase a greater efficiency of separation will occur [3]. Analysis of efficiency first depends on binary mole fractions in equilibrium. In which liquid mole fractions may be found via titration and in turn the equilibrium constant.

$$y_i = K_i(x_i)$$

y_i = mole fraction of solvent in extract phase

x_i = mole fraction of solvent initial feed phase

K_i = equilibrium constant of solvent

A McCabe Thiele diagram may then be constructed under equilibrium to correlate an operating line to stages. Use of the operating line to construct stages of separation are valid under the following conditions of the system: isothermal, isobaric, and heat of mixing is negligible [4]. Furthermore, mole fractions provide information to establish the slope of the operating line. From titration mole fractions may be set as points (x_n, y_{n+1}) for the operating line to meet and connect the total stages needed in the system. Lastly, Murphree’s efficiency may be associated to extract and raffinate mole fractions by $E_{ME,s}$ and E_{MR} respectively.

$$E_{ME,s} = \frac{Y_{s+1} - Y_s}{Y_{s+1}^* - Y_s} \quad E_{MR,s} = \frac{X_s - X_{s+1}}{X_s - X_{s+1}^*}$$

$E_{ME,s}$ = Murphree efficiency in the extract phase for stage s (%)

Y_{s+1}^* = composition of extract phase in equilibrium with raffinate phase

$E_{MR,s}$ = Murphree efficiency in the raffinate phase for stage s (%)

X_{s+1}^* = composition of raffinate phase in equilibrium with extract phase

It is important to note that the Murphree efficiency is valid upon different agitation speeds of mixing. In fact, different agitation speeds are used to produce different mole fractions per stage and in turn, a different efficiency. The flow rates of extract and feed remained constant due to little to no interactions between phases as immiscible liquids. Hence, the overall efficiency of the system is dependent upon number of stages immiscible liquid -liquid phases.

$$E_o = \frac{n_T}{n_A}$$

3

n_T = Total number of theoretical stages from McCabe – Thiele

n_A = Actual number of stages

4. Results (15 pts)

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In this experiment, 25 gallons of white oil was mixed with acetic acid such that the concentration of the acetic acid is 0.5 wt %. After adding the acetic acid into the tank, a sample was drawn from the tank. The sample was then titrated using 0.1 N NaOH solution to determine the initial concentration of the acetic acid. After the mixture is uniformly mixed into the tank, the water (solvent) flowrate is adjusted to be equal to 0.7 GPM and the oil (feed) flowrate was adjusted to be equal to 0.4 GPM such that the ratio of solvent to feed ratio equals 2. Once the steady state is reached, 20 mL samples of the oil and water phase were drawn from each stage at 3 different agitator speeds. For example, for the agitator speed of 400 RPM a total of six samples were collected. The similar approach was taken for the other two agitator speeds i.e. 600 RPM and 900 RPM. Similarly, to determine the concentration of the acetic acid in the two phases at each stage titrations were performed. The results of the titrations are shown in Tables 5-6, Appendix I.

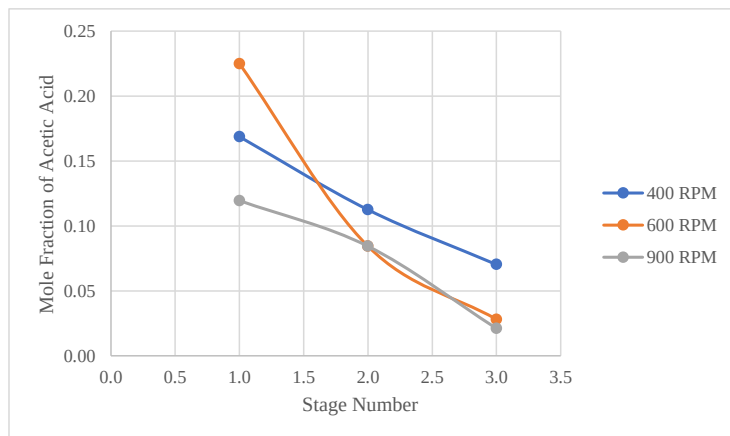


Figure 1 – Mole fraction of acetic acid versus stage number for the oil phase at various agitator speeds

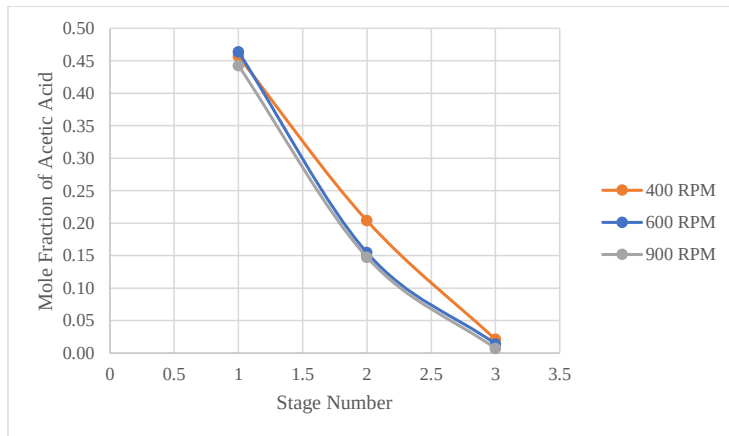


Figure 2 – Mole fraction of acetic acid versus stage number for the water phase at various agitator speeds

Figure 1 and 2 show the concentration of acetic acid as a function of the number of stages at three different agitator speeds. It was observed that the concentration of acetic acid in water phase increases as compared to the concentration of acetic acid in oil since more acetic acid is extracted out of the oil by water as we go higher in the stage level. Therefore, higher concentration of acetic acid was expected at higher stages. Also, it can be observed from the graphs that the concentration of the acetic acid decreases with increase in agitator speeds.

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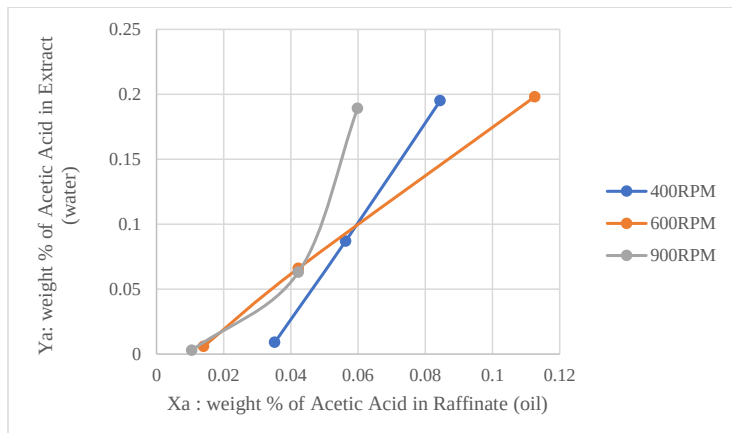


Figure 3 – Ya: weight % of Acetic Acid in Extract vs Xa: weight % of Acetic Acid in Raffinate

Figure 3 shows the relationship between the concentration of acetic acid in the extract (water/ Y_a) versus the concentration of acetic acid in the raffinate (oil/ X_a) at varying speeds of an agitator. As the amount of acetic acid in the raffinate Y_a increases, the amount that is being recovered in an extract X_a decreases. The plot shown in Figure 3 shows the similar behavior when the agitator speed is set to higher speed.

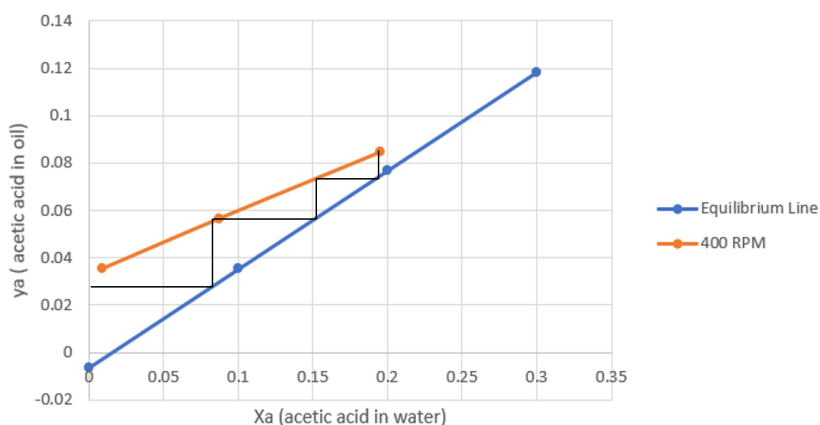


Figure 4 – McCabe Thiele Diagram for 400 RPM

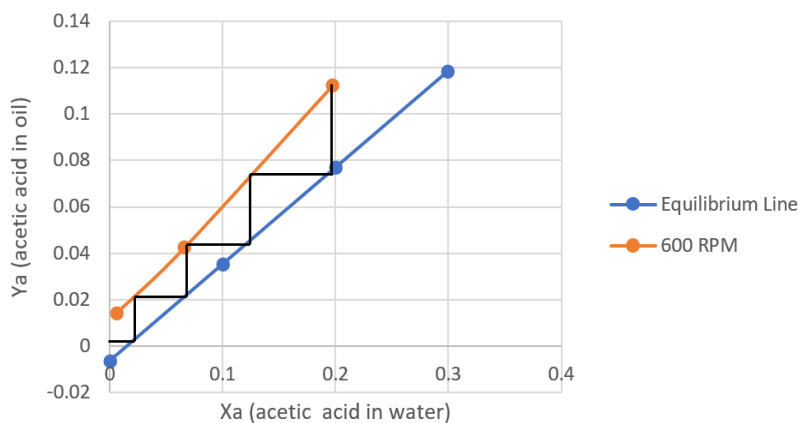


Figure 5 – McCabe Thiele Diagram for 600 RPM

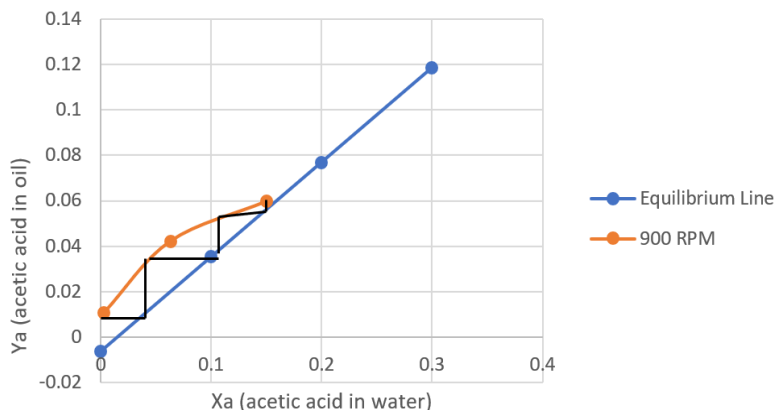


Figure 6 – McCabe Thiele Diagram for 900 RPM

Figure 4 through **Figure 6** shows the McCabe Thiele analysis performed in order to calculate the number of theoretical stages at various agitator speeds. The equilibrium data was provided by Dr. Zdunek. To determine the slope of the operating line, the ratio of the extract to raffinate (E/R) was taken. The slopes that we found were equal to 1.607, 1.465, and 1.930 for 400, 600, and 900 RPM agitator speeds respectively. Furthermore, using the staircase procedure, theoretical plates were calculated, and they were found to be 2.6, 3.2, and 2.8 for 400 RPM, 600 RPM, and 900 RPM agitator speeds respectively. The results for the theoretical plates are shown in **Figures 4, 5, and 6** for 400 RPM, 600 RPM, and 900 RPM respectively.

The **overall efficiency** is the ratio of the number of theoretical stages to the number of actual stages. As we increase the speed of an agitator, the overall efficiency of the extraction of acetic acid from the oil phase increases. The overall efficiencies were found to be 86.6 %, 106.7 %, and 93.3 % for 400, 600, and 900 RPM. The discrepancies for 600 RPM efficiency will be discussed in error analysis [section](#).

Murphee efficiency is calculated by taking a ratio of concentration difference before and after stage (actual) to the difference of the concentration at equilibrium. The calculated Murphee efficiency for an **extract phase** for **stage 1** is found to be 85.24 % and for the **raffinate phase** for **stage 1** is found to be 60.01 %.

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5. Discussion (15 pts)

The liquid-liquid extraction experiment consisted of three mixing-settling tanks, each with agitator system. The system was operated to extract the acetic acid initially mixed in white oil (raffinate) by using water (solvent). The extraction of acetic acid was observed by measuring the concentration of acetic acid in both the oil and water in each stages (tank) while the tanks' agitators are stirred at a certain RPM (400, 600, and 900). Initially, the water circuit was turned on, brought to a steady state and simultaneously the 0.5 wt% of acetic acid mixture was prepared by adding 387mL of acetic acid to 25 gallons of oil in the oil storage tank. It was mixed by pumping the mixture in closed circuit. The concentration of acetic acid present initially in the tank was measured by titrating a 20 mL of mixture sample with 0.01 N of NaOH. The extraction in the system was carried out by making the oil mixture flow counter current to the water flow in the three tanks i.e. the water was flowing from tank 3 to tank 1 while the oil mixture was flowing from tank 1 to tank 3 to the second storage tank. The water flow was set at 0.70 GPM while the oil flow was set at 0.40 GPM. The speeds of the agitators were set for three different trials i.e. 400, 600 and 900 RPM. Before taking samples for titration at differing agitation speeds, the system needed to reach steady state. The steady state was observed when the white oil, which is less dense than water, formed a layer on top of water; hence, two layers are form as a result of separation. The samples taken for each agitator speed were ~20 mL and titrated with 0.01 N NaOH to calculate the concentration of acetic acid in both the oil and water in each stage. However, the volume of 0.01 N NaOH was very large (>45 mL) for the 20 mL sample, so the titrations were then carried out by 0.1 N NaOH.

White oil is immiscible in water while acetic acid is soluble in both water and white oil i.e. in both kinds of polar and nonpolar compounds. When the oil – acetic acid mixture meets the water in the first tank, some of the acetic acid gets transferred from the oil to the water. This happens as a result of less concentration of acetic acid in water, and so the solute (acetic acid) travels from high concentration to low concentration. The transfer rate of acetic acid depends on both the area availability for contact and interfacial mass transfer coefficient. Both can be achieved to its maximum efficiency by the speed of the agitators. Hence, the speed of all three agitators play an important role in dispersion of acetic acid from the oil to the water [5]. The Figures 1 and 2 shows the acetic acid concentration in oil and water, respectively, at different agitator speeds i.e. 400, 600, 900 RPM vs the number of stages. It is observed in Figure 1 that the concentration of acetic acid in the oil decreases dramatically from stage 1 to stage 2 for 600 RPM. The other two speeds of the agitators, 400 and 900 RPM, showed similar trends to each other and decrease of the acetic acid concentration at a steady rate.

Figure 3 shows the wt% of acetic acid oil (raffinate) vs the wt% of acetic acid the water (solvent) for each speed of agitators. The feed of the mixture of oil and acetic acid comes into the first tank where the solvent i.e. water is leaving and vice-versa happens in the third tank. The concept of counter-flow system allows the maximum extraction from the raffinate phase to the solvent [6]. Additionally, in this case, acetic acid has more affinity to water in comparison to the white oil and so most of the acetic acid is being extracted in the first stage itself.

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pretty good discussion but you didn't explain the high Murphee efficiency for 600rpm

An equilibrium data of Water-White Oil-Acetic Acid was used to calculate number stages that are required by the system by using the McCabe-Thiele method. The Murphree stage and overall efficiencies were calculated by:

$$E_o = \frac{\text{number of the theoretical stages}}{\text{number of the actual stages}}$$

$$E_{ME} = \frac{Y_{s+1} - Y_s}{Y_{s+1}^* - Y_s} \quad E_{MR} = \frac{X_{s+1} - X_s}{X_{s+1}^* - X_s}$$

where,

E_o – overall efficiency

E_{ME} – Murphree efficiency of the extract phase

E_{MR} – Murphree efficiency of the raffinate phase

The theoretical stages obtained were 2.6, 3.2, and 2.8 for each agitator speed of 400, 600 and 900 RPM, respectively. The extraction of acetic acid from the oil requires longer time and hence, it requires continuous stirring by the agitators in each tank to escalate the extraction. As the agitator speeds are increased, the minimum residence time is decreased and hence increases the overall efficiency of the system. An addition to this system can be done by introducing a reflux in the extractor. A reflux in this countercurrent extraction will enhance the separation of the acetic acid properly from the oil and make the dispersion more efficient [7].

6. References (10 pts)

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[6] Ziegler, Erich, and Herta Ziegler. *Flavourings: Production, Composition, Applications, Regulations*. Weinheim: Wiley-VCH, 1998. Print.

[7] McCabe, Warren, Julian Smith, and Peter Harriott. *Unit Operation of Chemical Engineering*. 5th ed. N.p.: McGraw-Hill, 1993. Print.

7. Appendix I: Data Tabulation/Graphs (10 pts)

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Table 1: General Data of Compounds Used

Measurements	Values
Volume White Oil (L)	94.64
Oil Density (kg/L)	0.853
Weight Mineral Oil (kg)	80.724
Volume Acetic Acid (L)	0.387
Wt% Acetic Acid in Oil	0.5
Flow Rate of Oil (gal/min)	0.448
Flow Rate of Water (gal/min)	0.784
NaOH Normality	0.1
MW NaOH (g/mol)	40
Concentration of NaOH (g/mL)	0.01
MW Water (g/mol)	18
MW Acetic Acid (g/mol)	60.05
Total Concentration of Acetic Acid (mol/L) [Ct]	0.0711
Density Acetic Acid	1.049

Table 2: Recovered Mole Fraction of Acetic Acid

Agitator Speed (RPM)	Stage Number	Mole Fraction of Acetic Acid	
		Extract (water)	Raffinate (oil)
400	1	0.4568	0.1687
	2	0.2038	0.1124
	3	0.0211	0.0703
600	1	0.4638	0.2249
	2	0.1546	0.0843
	3	0.0141	0.0281
900	1	0.4428	0.1195
	2	0.1476	0.0843
	3	0.0070	0.0211

Table 3: Calculated Equilibrium Constant

Agitator Speed (RPM)	Stage Number	E/R	Average K
400	1	2.708	1.607
	2	1.813	
	3	0.300	
600	1	2.063	1.465
	2	1.833	
	3	0.500	
900	1	3.706	1.930
	2	1.750	
	3	0.333	

Table 4: Overall Percent Efficiency

Agitator Speed (RPM)	# Theoretical Plates	Overall Efficiency (%)
400	2.6	86.6
600	3.2	106.7
900	2.8	93.3

Table 5: 400 RPM Raffinate & Solvent Data

	Raffinate Stages -oil		
	1	2	3
Stream (Stage #)	X1	X2	X3
Volume NaOH (mL)	2.4000	1.6000	1.0000
Amount of OH [mol]	0.0002	0.0002	0.0001
Acetic Acid Present [mol]	0.0002	0.0002	0.0001
Molar Concentration of Acid [mol/L]	0.0120	0.0080	0.0050
Mass of Acetic Acid Present [g]	0.014412	0.009608	0.006005
Mass of Oil Present [g]	17.04828	17.05219	17.055117
Weight Fraction of Acetic Acid in Oil	0.0008	0.0006	0.0004
Weight %	0.084465	0.056313	0.035196982
Mole Fraction of acetic acid in the Raffinate (Fr1)	0.1687	0.1124	0.0703
	Solvent Stages		
Stream (Stage #)	Y1	Y2	Y3
Volume NaOH (mL)	6.5000	2.9000	0.3000
Amount of OH [mol]	0.0007	0.0003	0.00003
Acetic Acid Present [mol]	0.0007	0.0003	0.00003
Molar Concentration of Acid [mol/L]	0.0325	0.0145	0.0015
Mass of Acetic Acid Present [g]	0.039033	0.017415	0.0018015
Mass of water Present [g]	19.96279	19.9834	19.99828265
Weight Fraction of Acetic Acid in water	0.001951	0.000871	9.00746E-05
Weight %	0.195145	0.087069	0.009007462
Mole Fraction of acetic acid in the extract (Fe1)	0.4568	0.2038	0.0211
Ratio of Extract to Raffinate Concentration (E/R)	2.708333	1.8125	0.300
k (Average E/R)	1.606944444		

Table 6: 600 RPM Raffinate & Solvent Data

	Raffinate Stages		
	1	2	3
Stream (Stage #)	X1	X2	X3
Volume NaOH (mL)	3.2000	1.2000	0.4000
Amount of OH [mol]	0.0003	0.0001	0.0000
Acetic Acid Present [mol]	0.0003	0.0001	0.0000
Molar Concentration of Acid Acid [mol/L]	0.0160	0.0060	0.0020
Mass of Acetic Acid Present [g]	0.019216	0.007206	0.002402
Mass of Oil Present [g]	17.04437441	17.0541404	17.0580468
Weight Fraction of Acetic Acid in Oil	0.0011	0.0004	0.0001
Weight %	0.1126	0.0422	0.0141
Mole Fraction of acetic acid in the Raffinate (Fr1)	0.2249	0.0843	0.0281
	Solvent Stages		
Stream (Stage #)	Y1	Y2	Y3
Volume NaOH (mL)	6.6000	2.2000	0.2000
Amount of OH [mol]	0.0007	0.0002	0.00002
Acetic Acid Present [mol]	0.0007	0.0002	0.00002
Molar Concentration of Acid Acid [mol/L]	0.0330	0.0110	0.0010
Mass of Acetic Acid Present [g]	0.039633	0.013211	0.001201
Mass of water Present [g]	19.9622183	19.9874061	19.9988551
Weight Fraction of Acetic Acid in water	0.001981467	0.00066053	6.00498E-05
Weight %	0.198146659	0.066052962	0.006004983
Mole Fraction of acetic acid in the extract (Fe1)	0.4638	0.1546	0.0141
Ratio of Extract to Raffinate Concentration (E/R)	2.0625	1.833333333	0.500
K (Average E/R)	1.465277778		

Table 7: 900 RPM Raffinate & Solvent Data

	Raffinate Stages		
	1	2	3
Stream (Stage #)	X1	X2	X3
Volume NaOH (mL)	1.7000	1.2000	0.3000
Amount of OH [mol]	0.0002	0.0001	0.0000
Acetic Acid Present [mol]	0.0002	0.0001	0.0000
Molar Concentration of Acid Acid [mol/L]	0.0085	0.0060	0.0015
Mass of Acetic Acid Present [g]	0.010209	0.007206	0.0018015
Mass of Oil Present [g]	17.0517	17.05414	17.0585351
Weight Fraction of Acetic Acid in Oil	0.0006	0.0004	0.0001
Weight %	0.0598	0.0422	0.0106
Mole Fraction of acetic acid in the Raffinate (Fr1)	0.1195	0.0843	0.0211
	Solvent Stages		
Stream (Stage #)	Y1	Y2	Y3
Volume NaOH (mL)	6.3000	2.1000	0.1000
Amount of OH [mol]	0.0006	0.0002	0.00001
Acetic Acid Present [mol]	0.0006	0.0002	0.00001
Molar Concentration of Acid Acid [mol/L]	0.0315	0.0105	0.0005
Mass of Acetic Acid Present [g]	0.037832	0.012611	0.0006005
Mass of water Present [g]	19.96394	19.98798	19.99942755
Weight Fraction of Acetic Acid in water	0.001891	0.000631	3.0025E-05
Weight %	0.189141	0.063051	0.003002496
Mole Fraction of acetic acid in the extract (Fe1)	0.4428	0.1476	0.0070
Ratio of Extract to Raffinate Concentration (E/R)	3.705882	1.75	0.333
k (Average E/R)	1.929738562		

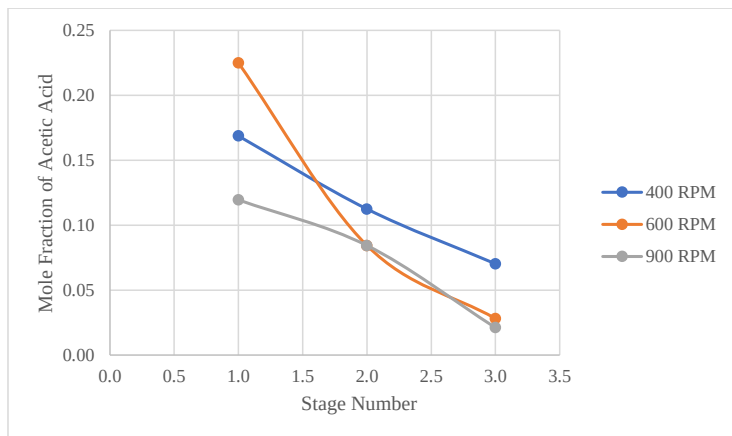


Figure 7 – Mole fraction of acetic acid versus stage number for the oil phase at various agitator speeds

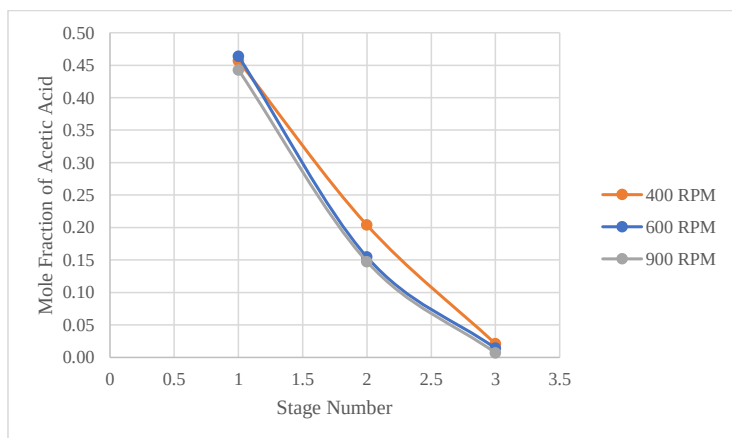


Figure 8 – Mole fraction of acetic acid versus stage number for the water phase at various agitator speeds

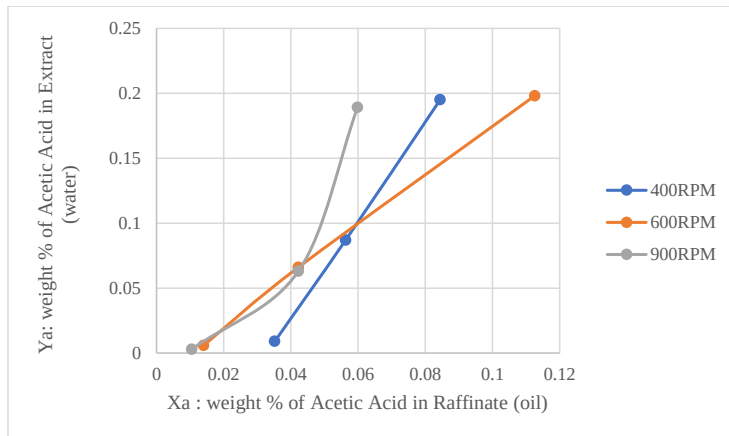


Figure 9 – Ya: weight % of Acetic Acid in Extract vs Xa: weight % of Acetic Acid in Raffinate

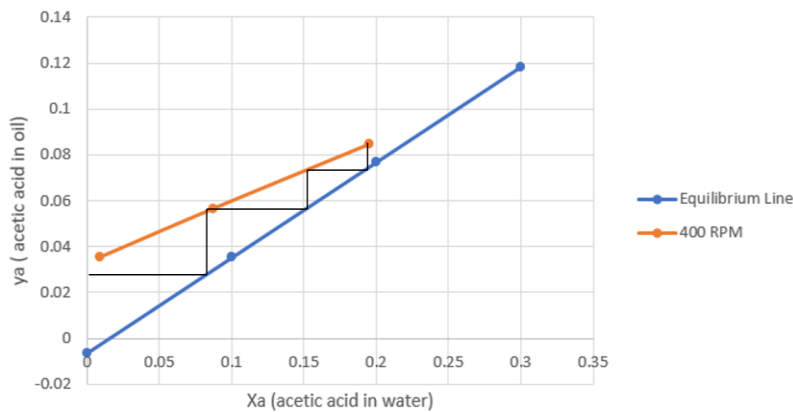


Figure 10 – McCabe Thiele Diagram for 400 RPM

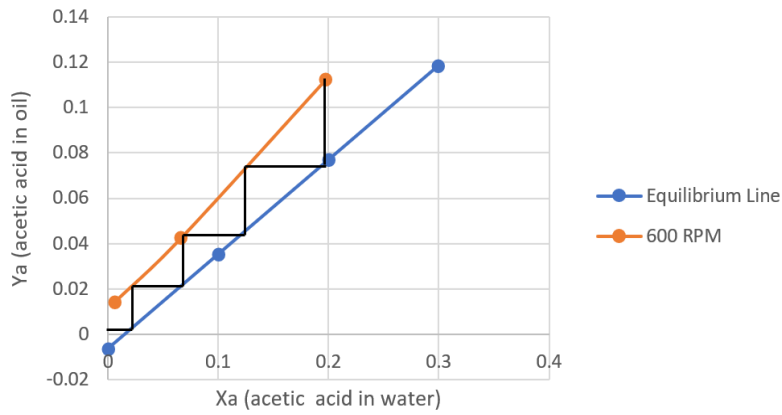


Figure 11 – McCabe Thiele Diagram for 600 RPM

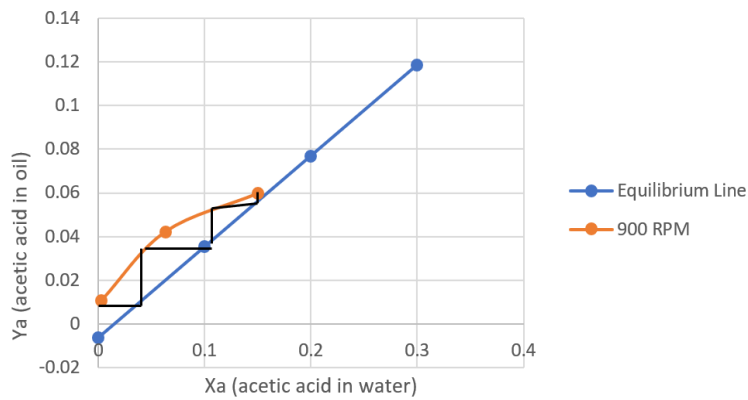


Figure 12 – McCabe Thiele Diagram for 900 RPM

8. Appendix II: Error Analysis (10 pts)

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Among experimentation of liquid-liquid extraction both systematic and random errors occurred. Initially, the prevalent error analyzes was with the flowmeters. As the equipment is a bit old and run down the valve for the flows would continuously move and therefore change the flowrates to not be constant. To account for this error a supervisor was assigned to maintain the flowrates constant by re-adjusting the valve as it fluctuated with time. It is noted that a earlier lab group operated the equipment for the same experiment. Although the extract tank and stages were initially drained the feed tank remained full. Some residues from draining the earlier lab experiment remained. The effect of the remains in the stages were not very prevalent in the data. Rather, titration errors may have produced greater error.

Each trial 20 mL sample was added 5 drops of phenolphthalein. As titration proceeded among two supervisors an established indication of “pink” did not occur until a third way through titrations. Qualitatively the pink tint did not vary too much, but more accurate volumes of NaOH may have been produced if the tint was established beforehand. Yet, the titration error of overshooting did not affect the data too much as seen in mole fractions among the established McCabe Thiele graphs. Instrumental errors did occur, and the uncertainty was noted for as seen below.

Table 8: Uncertainty Effects

Apparatus	Uncertainty	Error affect
10 mL pipette	$\pm 2\%$	transferring 20 mL of extract & raffinate volume
5 mL pipette	± 0.1 mL	volume of 5 drops of phenolphthalein delivered
Flowmeter	± 0.01 GPM	flow rate of oil and water into system
50 mL Erlenmeyer flask	$\pm 5\%$	holding trial volumes for titration
Agitators	± 5.0 RPM	mixing oil and water with acetic acid

Overall, the errors that occurred in the procedure and data of the lab did vary but did not vary to an extent that would provide inaccurate readings. The standard deviations for theoretical stages and efficiencies per 3 agitator speeds* are 0.31 & 10.23 respectively. As well as the percent error of theoretical stages to actual stages (3) is less than 15%

Table 9: Stages Percent Error.

Agitator Speed (RPM)	Percent Error of Stages
400	13.33
600	6.67
900	6.67

9. Appendix III: Sample Calculations (10 pts)

Commented [PTH14]: 10/10

Since this was the first time the white oil was used for the experiment, therefore the initial concentration of acetic acid in the white oil is zero.

- **Normality of NaOH**

$$N = \frac{m_{NaOH}}{\text{Volume of solution}} * \frac{1}{Eq}$$

Where,

N = normality of NaOH

m = mass of NaOH (since the solution used was N/10, m=4g)

Eq = grams equivalent weight, in the NaOH case is 40.00g/eq

$$N = \frac{4}{1} * \frac{1}{40} = 0.1$$

- **Amount of Acetic Acid to be Added**

Assuming density of acetic acid as 1.049g/mL [1M] and density of Keller-Heartt white oil as 0.853g/mL

$$25 \text{ gal of oil} * \left(\frac{3.785 \text{ L of oil}}{1 \text{ gal of oil}} \right) * \left(\frac{0.853 \text{ kg}}{1 \text{ L of oil}} \right) = 80.7151 \text{ kg of oil}$$

$$0.005 = \frac{\text{amount of acetic acid}}{\text{amount of acetic acid} + 80.7151} = 0.4056 \text{ kg of acetic acid}$$

$$0.4056 \text{ kg of acetic acid} * \left(\frac{1 \text{ L of acetic acid}}{1.049 \text{ kg}} \right) = 0.387 \text{ L of acetic acid}$$

The amount of acetic acid to be added is 0.387L or 387mL into the white oil solution.

- **Calculate the total molar concentration of acetic acid.**

$$M_{CT} = \frac{\frac{\text{Volume of acetic acid} * \text{density of acetic acid}}{\text{molecular weight of acetic acid}}}{\text{volume of mixture}}$$

Where

M_{CT} = total molar concentration of acetic acid (mol/L)

$$M_{CT} = \frac{\frac{387 \text{ mL} \times 1.049 \frac{\text{g}}{\text{mL}}}{60.05 \frac{\text{g}}{\text{mol}}}}{95.022 \text{ L}} = 0.0711 \text{ mol/L}$$

The following sample calculations were based on the agitator speed of 400 RPM and from the first storage tank.

- **Amount of acetic acid present in the water**

$$\begin{aligned} \text{amount of OH} &= \text{amount of acetic acid} \\ \text{Amount of acetic acid} &= \text{Volume of NaOH} * \text{Normality of NaOH} \\ \text{Amount of acetic acid} &= \frac{6.5 \text{ L}}{1000} * 0.1 \frac{\text{mol}}{\text{L}} = 0.00065 \text{ mols of acetic acid} \end{aligned}$$

- **Concentration of acetic acid in the water phase.**

$$M_{o,i} = \frac{\text{Amount of Acetic Acid}}{\text{Volume of the sample}}$$

Where

$M_{o,i}$ = molar concentration of acetic acid in oil phase at stage i (mol/L)

$$M_{o,1} = \frac{0.00065}{0.020 \text{ L}} = 0.0325 \text{ mol/L}$$

- **Mass of acetic acid present in the sample of water**

$$\begin{aligned} \text{Mass of acetic acid} &= \text{mols of acetic acid} * \text{molecular weight of acetic acid} \\ \text{Mass of acetic acid} &= 0.00065 \text{ mols} * 60.05 \frac{\text{g}}{\text{mol}} = 0.0390 \text{ g of acetic acid} \end{aligned}$$

- **Mass of water present in the sample of solvent**

$$\begin{aligned} \text{Mass of water} &= \left(\text{volume of the sample} - \frac{\text{mass of acetic acid}}{\text{density of acetic acid}} \right) \\ \text{Mass of water} &= \left(20 \text{ mL} - \frac{0.0390 \text{ g}}{1.049 \frac{\text{g}}{\text{mL}}} \right) = 19.96 \text{ g of water} \end{aligned}$$

- **Weight fraction of acetic acid in the solvent**

$$\begin{aligned} \text{weight fraction of acetic acid} &= \frac{\text{mass of acetic acid}}{\text{mass of acetic acid} + \text{mass of the water}} \\ \text{weight fraction of acetic acid} &= \frac{0.0390 \text{ g}}{0.0390 \text{ g} + 19.92 \text{ g}} = 0.0019 \end{aligned}$$

- **Amount of acetic acid present in the oil**

$$\begin{aligned} \text{amount of OH} &= \text{amount of acetic acid} \\ \text{Amount of acetic acid} &= \text{Volume of NaOH} * \text{Normality of NaOH} \\ \text{Amount of acetic acid} &= \frac{2.4 \text{ L}}{1000} * 0.1 \frac{\text{mol}}{\text{L}} = 0.00024 \text{ mols of acetic acid} \end{aligned}$$

- **Concentration of acetic acid in the oil phase.**

$$M_{o,i} = \frac{\text{Amount of Acetic Acid}}{\text{Volume of the sample}}$$

Where

$M_{o,i}$ = molar concentration of acetic acid in oil phase at stage i (mol/L)

$$M_{o,1} = \frac{0.00024}{0.020\text{ L}} = 0.0120\text{ mol/L}$$

- **Mass of acetic acid present in the sample**

*Mass of acetic acid = mols of acetic acid * molecular weight of acetic acid*

$$\text{Mass of acetic acid} = 0.00024\text{ mols} * 60.05 \frac{\text{g}}{\text{mol}} = 0.0144\text{g of acetic acid}$$

- **Mass of oil present in the sample of raffinate**

*Mass of oil = $\left(\text{volume of the sample} - \frac{\text{mass of acetic acid}}{\text{density of acetic acid}} \right) * \text{density of oil}$*

$$\text{Mass of oil} = \left(20\text{mL} - \frac{0.0144\text{g}}{1.049 \frac{\text{g}}{\text{mL}}} \right) * 0.853 \frac{\text{g}}{\text{mL}} = 17.05\text{g of oil}$$

- **Weight fraction of acetic acid in the raffinate**

$$\text{weight fraction of acetic acid} = \frac{\text{mass of acetic acid}}{\text{mass of acetic acid} + \text{mass of the oil}}$$

$$\text{weight fraction of acetic acid} = \frac{0.0144\text{g}}{0.0144\text{g} + 17.05\text{g}} = 0.0008$$

- **Mole percentage of acetic acid in the solvent phase.**

$$F_{E,i} = \frac{C_{w,i}}{C_T} \times 100$$

Where

$F_{E,i}$ = mole fraction in the extract phase at stage i

$$F_{E,1} = \frac{0.0325 \frac{\text{mol}}{\text{L}}}{0.0711 \frac{\text{mol}}{\text{L}}} \times 100 = 45.68\text{ mol}\%$$

- **Mole Percentage of acetic acid in the raffinate phase.**

$$F_{R,i} = \frac{C_{o,i}}{C_T} \times 100$$

Where

$F_{R,i}$ = mole fraction in the raffinate phase at stage i

$$F_{R,1} = \frac{0.012 \frac{\text{mol}}{\text{L}}}{0.0711 \frac{\text{mol}}{\text{L}}} \times 100 = 16.87 \text{ mol\%}$$

- **Ratio of extract to raffinate concentrations (E/R).**

$$\left(\frac{E}{R}\right)_1 = \frac{45.68}{16.87} = 2.70$$

- **Average E/R for each agitator speed.**

$$K_{AS} = \frac{1}{n} \sum_{i=1}^n \left(\frac{E}{R}\right)_i$$

Where

K_{AS} = average E/R (or slope) for the designated agitator speed

n = total number of stages

$$K_{400} = \frac{1}{3} \left[\left(\frac{E}{R}\right)_1 + \left(\frac{E}{R}\right)_2 + \left(\frac{E}{R}\right)_3 \right] = \frac{1}{3} (2.70 + 1.81 + 0.300) = 1.61$$

- **Murphree efficiency in the extract phase.**

$$E_{ME,s} = \frac{Y_{s+1} - Y_s}{Y_{s+1}^* - Y_s}$$

Where

$E_{ME,s}$ = Murphree efficiency in the extract phase for stage s (%)

Y_{s+1}^* = Composition of extract phase in equilibrium with raffinate phase

$$E_{ME,1} = \frac{0.2038 - 0.4568}{0.16 - 0.4568} * 100 = 85.24\%$$

- **Murphree efficiency in the raffinate phase.**

$$E_{MR,s} = \frac{X_s - X_{s+1}}{X_s - X_{s+1}^*}$$

Where

$E_{MR,s}$ = Murphree efficiency in the raffinate phase for stage s (%)

X_{s+1}^* = composition of raffinate phase in equilibrium with extract phase

$$E_{MR,1} = \frac{0.1687 - 0.1124}{0.1687 - 0.075} * 100 = 60.01\%$$

- **Overall efficiency.**

$$E_o = \frac{n_{\text{Total number of theoretical trays from McCabe-Thiele}}}{n_{\text{actual number of trays}}} * 100$$

Where

E_o = overall efficiency (%)

$$E_o = \frac{2.6}{3} * 100 = 86.6\%$$

10. Appendix IV: Individual Team Contributions

NAME: Huda Usman

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	N/A
Pre-Lab Editing	1	Edited mine and other's parts of the prelab
Experimental Objective (Prelab)	0.3	Helped another member
Apparatus (prelab)	0.3	Wrote the apparatus with another team member
Materials & Supplies (prelab)	0.3	Wrote the supplies with another team member
Procedure (prelab)	0.30	Wrote the procedure with another team member
Project Safety Evaluation (prelab)	0.3	Wrote the safety evaluation with another team member
Final Lab Editing	2	Worked with all the team members
Executive Summary (final lab)	0.15	Helped another member
Introduction (Final lab)	0	Assigned to another member
Theory (Final Lab)	0	Assigned to another member
Results (final lab)	6	Worked on it
Discussion (final lab)	0	Assigned to another member
References (final and/or prelab)	0.30	Wrote the references with another team member
Data Tabulation/Graphs (final)	0	Assigned to another member
Error Analysis (final lab)	0	Assigned to another member
Sample Calculations (final Lab)	0	Assigned to another member
Power Point Presentation	1	Helped writing other team members
Total	21.15	

NAME: Nisha Patel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	N/A
Pre-Lab Editing	1	Edited mine and other's parts of the prelab
Experimental Objective (Prelab)	0.3	Helped another member
Apparatus (prelab)	0.3	Wrote the apparatus with another team member
Materials & Supplies (prelab)	0.3	Wrote the supplies with another team member
Procedure (prelab)	0.30	Wrote the procedure with another team member
Project Safety Evaluation (prelab)	0.3	Wrote the safety evaluation with another team member
Final Lab Editing	2	Worked with all the team members
Executive Summary (final lab)	0.15	Helped another member
Introduction (Final lab)	0	Assigned to another member
Theory (Final Lab)	0	Assigned to another member
Results (final lab)	0	Assigned to another member
Discussion (final lab)	6	Worked on it
References (final and/or prelab)	0.30	Wrote the references with another team member
Data Tabulation/Graphs (final)	0	Assigned to another member
Error Analysis (final lab)	0	Assigned to another member
Sample Calculations (final Lab)	0	Assigned to another member
Power Point Presentation	1	Helped writing other team members
Total	21.15	

NAME: Daniela Estarita

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Worked on titration
Pre-Lab Editing	0.5	Edited assigned part
Experimental Objective (Prelab)	0	n/a
Apparatus (prelab)	1	Worked on & labeled pictures
Materials & Supplies (prelab)	1	Worked on
Procedure (prelab)	0	n/a
Project Safety Evaluation (prelab)	0	n/a
Final Lab Editing	1	Edited assigned part
Executive Summary (final lab)	0	n/a
Introduction (Final lab)	0	n/a
Theory (Final Lab)	4	Worked on theory
Results (final lab)	0	n/a
Discussion (final lab)	0	n/a
References (final and/or prelab)	1	Found creditable sources for theory
Data Tabulation/Graphs (final)	1	Error tables
Error Analysis (final lab)	2	Worked on qualitative and quantitative errors
Sample Calculations (final Lab)	0	n/a
Power Point Presentation	1.5	Worked on theory & editing
Total	21	

NAME: Jasmine Villegas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Took Samples
Pre-Lab Editing	1	Edited all
Experimental Objective (Prelab)	1	Assigned task
Apparatus (prelab)	0	Assigned to others
Materials & Supplies (prelab)	0	Assigned to others
Procedure (prelab)	1	Edited
Project Safety Evaluation (prelab)	1	Edited
Final Lab Editing	3	Edited all
Executive Summary (final lab)	1	Assigned task
Introduction (Final lab)	0.5	Edited
Theory (Final Lab)	0.5	Edited
Results (final lab)	0.5	Edited
Discussion (final lab)	0.5	Edited
References (final and/or prelab)	0.5	Edited
Data Tabulation/Graphs (final)	2	Added graphs
Error Analysis (final lab)	0	Assigned to others
Sample Calculations (final Lab)	0	Assigned to others
Power Point Presentation	2	Added safety/problems/edited
Total	23.5	

NAME: Mariana St. Peter

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	Titration
Pre-Lab Editing	1	Edited all
Experimental Objective (Prelab)	0	Assigned to others
Apparatus (prelab)	0	Assigned to others
Materials & Supplies (prelab)	1	Assigned task
Procedure (prelab)	2	Assigned task
Project Safety Evaluation (prelab)	1	Assigned task
Final Lab Editing	1	Helped others
Executive Summary (final lab)	1	Helped others
Introduction (Final lab)	0.5	Edited
Theory (Final Lab)	0.5	Edited
Results (final lab)	0.5	Edited
Discussion (final lab)	0.5	Edited
References (final and/or prelab)	0.5	Edited
Data Tabulation/Graphs (final)	2	Composed graphs
Error Analysis (final lab)	0	Assigned to others
Sample Calculations (final Lab)	4	Completed all calculations
Power Point Presentation	1	Edited
Total	24.5	